Dr. Kewei Sha

University of North Texas
College of AI & Advanced Analytics
3940 N Elm St
Denton, TX 76207

Office: Discovery Park E295A
Phone: 940-565-3271
Email: kewei.sha@unt.edu
Web: https://comersha.github.io

Research Interests

Applied AI, Edge Computing, Physical AI, Trustworthy AI, Explainable AI, AI Governance, Trustworthy AI, Intelligent Healthcare Systems, and Internet of Things

Employment

09/2026-	Associate Chair, Department of Data Science University of North Texas (UNT)
03/2026-	Director, Data Science PhD Program University of North Texas (UNT)
09/2024-	Director, Data Science Graduate Program University of North Texas (UNT)
09/2023-	Associate Professor of Data Science University of North Texas (UNT)
09/2023-	Director, Secure, Reliable, and Intelligent Systems Laboratory University of North Texas (UNT)
09/2025-	Affiliated Professor of Information Science University of North Texas (UNT)
11/2025-	Affiliated Professor of Computer Science and Engineering University of North Texas (UNT)
01/2025-08/2025	Director, Data Science Undergraduate Program University of North Texas (UNT)
06/2023-08/2023	Professor of Computer Science University of Houston–Clear Lake (UHCL)
09/2019-06/2023	Associate Professor of Computer Science University of Houston–Clear Lake (UHCL)
09/2015-06/2023	Associate Director, Cyber Security Institute University of Houston–Clear Lake (UHCL)
09/2015-08/2019	Assistant Professor of Computer Science University of Houston–Clear Lake (UHCL)
07/2013-08/2015	Chair, Department of Technology and Software Engineering Oklahoma City University
07/2015-08/2015	Associate Professor of Computer Science Oklahoma City University
07/2011-06/2015	Assistant Professor of Computer Science Oklahoma City University
08/2008-06/2011	Visiting Assistant Professor of Computer Science Oklahoma City University

Education

Ph.D. in Computer Science, 09/2003-12/2008
Wayne State University, Detroit, MI

M.S. in Computer Science, 09/2003-12/2006
Wayne State University, Detroit, MI

B.S. in Computer Science, 09/1997-07/2001
East China University of Science and Technology, Shanghai, China

Honors and Awards

- **Best Paper Award**, IEEE/ACM CHASE 2026, 2026
- **Best Paper Award**, IEEE MOST 2026, 2026
- Outstanding Service Award from IEEE Healthcom 2026, 2026
- Leadership Fellow, University of North Texas (UNT), 2025-2026
- **Outstanding Paper Award**, Digital Communications and Networks, 2025
- Outstanding Service Award from IEEE/ACM CHASE 2025, 2025
- Above the Call Award Runners-Up, College of Information, UNT, 2025
- Departmental Service and Leadership Award, Department of Data Science, UNT, 2025
- Recognition of Service as General Chair of IEEE MOST 2024, 2024
- Recognition of Service as General Chair of IEEE/ACM SEC 2023, 2023
- University Faculty Fellowship Award, UHCL, 2022
- UHCL Anthony & Barbara Lekkos Endowment Faculty Fellowship, 2021
- Recognition of Service as General Chair of IEEE/ACM CHASE 2021 by ACM, 2021
- Recognition of Service as Program Chair of ACM/IEEE SEC 2020 by ACM, 2020
- UHCL President's Distinguished Research Award, 2019
- IEEE Outstanding Leadership Award from IEEE ICCCN 2019
- Paper "On security challenges and open issues in Internet of Things" is featured in Future Generation Computer Systems' Research Insights, 2019
- 2018 Albert Nelson Marquis Lifetime Achievement Award, 2018
- IEEE Outstanding Leadership Award 2018
- UHCL Piper Award Nominee, 2018
- Nominee of USERN Prize 2018
- UHCL Spotlight Researcher, 2018
- Elsevier Outstanding Reviewer, 2018
- IEEE Senior Member, 2016
- ACM Senior Member, 2016
- IEEE Outstanding Leadership Award 2015
- IEEE Outstanding Leadership Award 2014
- Recognition of Workshop Co-Chair from ICCCN 2013
- Distinguished Student Mentor of Meinders School of Business, OCU, 2012
- Fellow of the Faculty Teaching Academy at OCU, 2011
- Semi-Finalist in 2012 Oklahoma Governor's Business Plan Competition (as advisor), 2012
- The 3rd Place in 2011 Oklahoma Governor's Business Plan Competition (as advisor), 2011
- General Motor Corporations Academic Award, 2008

Awarded Funding

Total received at UNT: **$1,801,160**

- **Title: Excellence in Research: Trustworthy AI for Healthcare: Evidence-Grounded, Compliant, Human-in-the-Loop**
  Role: co-PI Agency: NSF Award#: 2602903
  Period: 10/2026-09/2029 Amount:$699,999
- **Title: Development and Testing of a Mechanism-Driven Dialectical Behavior Therapy Skills Coaching Powered by Generative AI: A Just-in-Time Adaptive Intervention to Reduce HIV and Substance Use Risks (Subaward)**
  Role: Senior Personnel Agency: NIH Award#: 1DP2DA066204-01
  Period: 09/2026-08/2029 Amount:$199,999
- **Title: RET Site: Teacher-Researcher Co-Design of Adaptive GenAI for Computational Thinking**
  Role: PI Agency: NSF Award#: 2601493
  Period: 05/2026-04/2029 Amount:$599,496
- **Title: An Edge-Based Approach to Robust Multi-Robot Systems in Dynamic Environments**
  Role: PI Agency: NSF Award#: 2505686
  Period: 03/2025-09/2027 Amount:$187,666
- **Title: Collaborative Research: Conference: NSF Computer and Information Science and Engineering Research Expansion Program (CISE MSI) 2024-2026 Aspiring Principal Investigators Workshop**
  Role: PI Agency: NSF Award#: 2437481
  Period: 08/2024-07/2026 Amount:$100,000
- **Title: Robust Generative AI Models for Safer Autonomous Vehicles**
  Role: co-PI Agency: NVIDIA Academic Grant
  Period: 10/2025-09/2026 Amount:$10,000
- **Title: Hosting an academic international conference (IEEE MOST 2024)**
  Role: PI Agency: UNT Global Venture Fund (GVF)
  Period: 11/2023-10/2024 Amount:$4,000

Total funds before moving to UNT: **$5,379,676** (at UHCL); $20,500 (at OCU)

- **Title: An Edge-Based Approach to Robust Multi-Robot Systems in Dynamic Environments**
  Role: PI Agency: NSF
  Period: 10/2022-09/2025 Amount: $234,982 Award #: 2240513
- **Title: An Edge-Based Approach to Robust Multi-Robot Systems in Dynamic Environment**
  Role: PI Agency: ASEE CyBR-MSI 2022
  Period: 05/2022-09/2022 Amount: $10,000
- **Title: UH ACPV Grant**
  Role: Senior Personnel Agency: UH
  Period: 06/2022-09/2022 Amount: $100,000
- **Title: NASA STTR: Blockchain for Model-Based System Engineering**
  Role: Senior Personnel Agency: NASA
  Period: 2021-2023 Amount: $750,000
- **Title: Recruiting and Supporting a Diverse Pool of Highly-Qualified STEM Teachers through Communities of Practice and Informal Science Internships**
  Role: co-PI Agency: NSF Noyce
  Period: 04/01/2022-03/31/2026 Amount: $1,199,969 Award #: 2151067

- **Title:** **Anthony & Barbara Lekkos Endowment Faculty Fellowship**
  Role: PI Agency: UHCL
  Period: 2021-2022 Amount: $10,000
- **Title:** **Building Capacity: Improving STEM Graduation Rates through Engaged Learning**
  Role: co-PI Agency: NSF HSI
  Period: 10/2019-09/2024 Amount: $2,499,930 Award#: 1928622
- **Title:** **Revamping a Traditional Computer Science Program to Meet Modern-day Cybersecurity Education Challenges**
  Role: co-PI Agency: NSF DGE
  Period: 09/2017-08/2020 Amount: $499,995 Award #: 1723596
- **Title:** **Creating Augmented Reality Platform for Treatment of Phobias\Anxiety with ASD Children**
  Role: PI (Sub-award) Agency: NSF IIP
  Period: 09/2019-08/2020 Amount: $43,800
- **Title:** **Student Travel Support for the IEEE SECON 2017 Conference**
  Role: PI Agency: NSF
  Period: 05/2017-04/2018 Amount: $10,000 Award #: 1734656
- **Title:** **SCH: Student Travel Support for IEEE/ACM CHASE 2017 Conference**
  Role: PI Agency: NSF
  Period: 07/2017-06/201 Amount: $15,000 Award #: 1720822
- **Title:** **Travel Award from the Sustainable Horizons Institute & Berkeley Lab**
  Role: PI Agency: Sustainable Horizons Institute
  Period: 2017 Amount: $1,200
- **Title:** **Faculty Research Support Funding (FRSF)**
  Role: PI Agency: UHCL
  Period: 07/2022-06/2023 Amount: $6,000
- **Title:** **Faculty Development Funding (FDF)**
  Role: PI Agency: UHCL
  Period: 09/2021-08/2022 Amount: $2,000
- **Title:** **Faculty Research Support Funding (FRSF)**
  Role: PI Agency: UHCL
  Period: 09/2019-08/2020 Amount: $5,900
- **Title:** **Faculty Development Funding (FDF)**
  Role: PI Agency: UHCL
  Period: 09/2018-08/2019 Amount: $2,000
- **Title:** **Faculty Development Funding (FDF)**
  Role: PI Agency: UHCL
  Period: 09/2017-08/2018 Amount: $2,000
- **Title:** **Faculty Development Funding (FDF)**
  Role: PI Agency: UHCL
  Period: 09/2016-08/2017 Amount: $2,000
- **Title:** **Faculty Research Support Funding (FRSF)**
  Role: PI Agency: UHCL
  Period: 09/2016-08/2017 Amount: $3,900
- **Title:** **Faculty Development Funding (FDF)**
  Role: PI Agency: UHCL
  Period: 09/2015-08/2016 Amount: $1,800
- **Title:** **Student Travel Support for IEEE ICCCN 2015 Conference**
  Role: PI Agency: NSF
  Period: 07/2015-06/2015 Amount: $10,000 Award#: 1550442
- **Title:** **NSF EPSCoR Oklahoma ROA+**
  Role: PI (sub-awarded) Agency: NSF
  Period: 05/2015-08/2015 Amount: $12,500

- **Title:** **Summer Undergraduate Research Grant**
  Role: PI Agency: OCU
  Period: 09/2014-08/2015 Amount: $5,000
- **Title:** **Student Professional Fund**
  Role: PI Agency: OCU
  Period: 09/2014-08/2015 Amount: $1,600
- **Title:** **SC'13 HPC Educator Travel Award**
  Role: PI Awardee Agency: SC Conference
  Period: 09/2013-12/2013 Amount: $2,000

Publications

(Corresponding author is marked with *)

Book & Book Chapters

1. Chen, H., **Sha, K.***, Zhou, Y., & Tu, F. (2027). Data quality: Models, measures, and tools. In *Springer Handbook of Data Engineering* (pp. 1013-1032). Springer. https://doi.org/10.1007/978-3-031-92372-2_55
2. **Sha, K.***, Cherukuri, K. S., & Chen, H. (2025). Privacy and bias in edge computing. In *Edge computing: Systems and applications* (pp. 223–252). IEEE. https://doi.org/10.1002/9781394285860.ch8
3. Peters, M. L., Yang, T. A., Wei, W., **Sha, K.,** & Davari, S. (2021). Examining the influence of participating in a cyber defense track on students' cybersecurity knowledge, awareness, and career choices. In *Advances in software engineering, education, and e-learning* (pp. 31–42). Springer. https://doi.org/10.1007/978-3-030-70873-3_3
4. **Sha, K.***, Striegel, A., & Song, M. (Eds.). (2017). *Advances in computer communications and networks: From green, mobile, pervasive networking to big data computing*. River Publishers.
5. **Sha, K.***, Striegel, A., & Song, M. (Eds.). (2017). *Security, privacy and reliability in computer communication and networks*. River Publishers.
6. Li, W., **Sha, K.**, & Zeadally, S. (2016). Routing in wireless sensor network–based cyber-physical systems. In *Cyber-physical system design with sensor networking technologies*. Springer. https://doi.org/10.1049/PBCE096E_ch7
7. **Sha, K.** (2010). *Data quality management in wireless sensor networks: A consistency model–based approach*. VDM Verlag Dr. Müller.
8. **Sha, K.,** Shi, W., & Sellamuthu, S. (2006). Load balanced query protocols for wireless sensor networks. In S. Phoha, T. F. La Porta, & C. Griffin (Eds.), *Sensor network operations*. Wiley–IEEE Press.

Journals Publications

1. Sattari, M., Zhou, Y., Chen, H., & **Sha, K.*** (2026). Modeling and Quantifying Redundancy in Multisource and Multimodal Autonomous Driving Data*, IEEE Transactions on Vehicular Technology (IEEE TVT)*. https://10.1109/TVT.2026.3745770
2. Zhou, Y., Li, R., Cleveland, A., Ding, J., **Sha, K**., Chen, H. (2026). Quality-aware multi-source data fusion and enhancement for Medical Concept Normalization using Large Language Models,

Journal of Biomedical Informatics, Volume 182, 2026,105093,https://doi.org/10.1016/j.jbi.2026.105093

3. Montazeri, S., Feng, Y., & **Sha, K.*** (2026). PublicAgent: Multi-agent design principles from an LLM-based open data analysis framework. *Digital Engineering.* https://doi.org/10.1016/j.dte.2026.100136
4. Wang, F., Li, D., Ma, B., Wang, X., Li, C., Qi, X., Zhang, Y., **Sha, K**., Chen, H., Yu, G. (2026), Selective Semantic State Editing for Private Edge-Cloud Inference in SLMs. *Journal of Cloud Computing.* https://doi.org/10.1186/s13677-026-00939-w
5. Cherukuri, K. S., **Sha, K**., & Huang, Z. (2026). Operationalizing federated learning for dynamic work zone safety. *Journal of Transportation Safety & Security*. 1–34. https://doi.org/10.1080/19439962.2026.2669126
6. Montazeri, S., Dargie, W., Feng, Y., & **Sha, K.*** (2026). Human activity recognition based on electrocardiogram data only. *Smart Health*. 10, Volume 40, 100652, https://doi.org/10.1016/j.smhl.2026.100652
7. Y. Zhou, H. Chen and **Sha, K.***, "A Novel Multilayer Task-Centric and Data Quality Framework for Autonomous Driving," in *IEEE Internet Computing*, vol. 30, no. 2, pp. 67-75, March-April 2026, https://doi.org/10.1109/MIC.2026.3665121
8. **Sha, K.***, Yue, K.-B., Wei, W., Wu, Y., Koduru, M., & Vuchuru, P. (2025). ConsortiumSec: Blockchain-based distributed security framework for consortium applications. *ACM Distributed Ledger Technologies: Research and Practice*, *4*(4), 1–19. https://doi.org/10.1145/3699965
9. Sun, H., Wang, Z., Chen, Y., Yu, Y., Wu, Y. &. **Sha, K.**, (2024). EdgeBrain: A game based collaborative computational task offloading framework for edge video analytics. *IEEE Transactions on Service Computing*, *5*, 2287–2303. https://doi.org/10.1109/TSC.2023.3342800
10. Sun, H., Zhang, X., Zhang, B., **Sha, K.**, & Shi, W. (2024). Optimal task offloading and trajectory planning algorithms for collaborative video analytics with UAV-assisted edge in disaster rescue. *IEEE Transactions on Vehicular Technology*, *73*(5), 6811–6828. https://doi.org/10.1109/TVT.2023.3344281
11. Ding, R., Xu, Y., Zhong, H., Cui, J., Ma, J., & Sha, K. (2023). Towards fully anonymous integrity checking and reliability authentication for cloud data sharing. *IEEE Transactions on Services Computing*, *16*(5), 3782–3795. https://doi.org/10.1109/TSC.2023.3270580
12. Sun, H., Chen, Y., **Sha, K.**, Huang, S., Wang, X., & Shi, W. (2023). A proactive on-demand content placement strategy in edge intelligent gateways. *IEEE Transactions on Parallel and Distributed Systems*, *34*(7), 2072–2090. https://doi.org/10.1109/TPDS.2023.3249797
13. Sun, H., Li, Q., **Sha, K.**, & Yu, Y. (2022). ElasticEdge: Edge computing-enabled intelligent video analytics framework for live video. *IEEE Internet of Things Journal*, *9*(22), 23031–23046. https://doi.org/10.1109/JIOT.2022.3187621
14. Sun, H., Yu, Y., **Sha, K.**, & Zhong, H. (2022). EdgeEye: A data-driven approach for optimal deployment of edge video analytics. *IEEE Internet of Things Journal*, *9*(19), 19273–19295. https://doi.org/10.1109/JIOT.2022.3166896
15. Sudhakar, S. R. V., Kayastha, N., & **Sha, K.*** (2022). OFES: Optimal feature evaluation and selection for multi-class classification. *Data & Knowledge Engineering*, *139*, Article 102007. https://doi.org/10.1016/j.datak.2022.102007
16. Sun, H., Zhang, B., Yu, Y., Zhang, X., **Sha, K.**, & Shi, W. (2022). FlexEdge: Dynamic task scheduling for a UAV-based on-demand mobile edge server. *IEEE Internet of Things Journal*, *9*(17), 15983–16005. https://doi.org/10.1109/JIOT.2022.3152447
17. Azeroual, O., Jha, M., Nikiforova, A., **Sha, K.**, Alsmirat, M., & Jha, S. (2022). A record linkage-based deduplication of research data: A DataCleaner extension. *Multimodal Technologies and Interaction*, *6*(4), Article 27. https://doi.org/10.3390/mti6040027
18. Li, Y., Yin, Y., Chen, X., Wan, J., Jia, G., & **Sha, K.** (2022). A secure dynamic mix zone pseudonym changing scheme based on traffic context prediction. *IEEE Transactions on Intelligent Transportation Systems*, *23*(7), 9492–9505. https://doi.org/10.1109/TITS.2021.3125744
19. Tang, J., Chen, Y., She, G., Xu, Y., **Sha, K.**, Wang, X., Wang, Y., & Hui, P. (2021). Identifying mis-configured author profiles on Google Scholar using deep learning. *Applied Sciences*, *11*(15), Article 6912. https://doi.org/10.3390/app11156912

20. Sudhakar, S. R. V., Kayastha, N., & **Sha, K.*** (2021). ActID: An efficient framework for activity sensor-based user identification. *Computers & Security*, *108*, Article 102319. https://doi.org/10.1016/j.cose.2021.102319
21. **Sha, K.***, Wei, W., Yang, T. A., & Davari, S. (2020). A survey of edge computing based designs for IoT security. *Digital Communications and Networks*, *6*(2), 195–202. **(Outstanding Paper Award)** https://doi.org/10.1016/j.dcan.2019.08.006
22. Sun, H., Yu, Y., **Sha, K.**, & Lou, B. (2019). mVideo: Edge computing enabled mobile video processing systems. *IEEE Access*, *7*, 11615–11623. https://doi.org/10.1109/ACCESS.2019.2963159
23. Li, Y., Dong, Z., **Sha, K.**, Jiang, C., Wan, J., & Wang, Y. (2019). TMO: Time domain outsourcing attribute-based encryption scheme for data acquisition in edge computing. *IEEE Access*, *7*, 40240–40257. https://doi.org/10.1109/ACCESS.2019.2907319
24. **Sha, K.***, Wei, W., Yang, T. A., Wang, Z., & Shi, W. (2018). On security challenges and open issues in Internet of Things. *Future Generation Computer Systems*, *83*, 326–337. https://doi.org/10.1016/j.future.2018.01.059
25. **Sha, K.***, Alatrash, N., & Wang, Z. (2017). A secure framework to read isolated smart grid devices. *IEEE Transactions on Smart Grid*, *8*(6), 2519–2531. https://doi.org/10.1109/TSG.2016.2526045
26. Basheer, A., & **Sha, K.***. (2017). Cluster-based quality-aware adaptive data compression for streaming data. *ACM Journal of Data and Information Quality*, *9*(1), Article 2. https://doi.org/10.1145/3122863
27. Du, J., Liu, S., Li, R., Yang, X., Liu, H., & **Sha, K.** (2017). Energy-efficient algorithm to construct the information potential field in WSNs. *IEEE Sensors Journal*, *17*(12), 3822–3831. https://doi.org/10.1109/JSEN.2017.2690338
28. **Sha, K.**, & Zeadally, S. (2015). Data quality in cyber-physical systems. *ACM Journal on Data and Information Quality*, *6*(2–3), Article 8. https://doi.org/10.1145/2740965
29. Wang, Z., **Sha, K.***, & Lv, W. (2013). Slight homomorphic signature for access controlling in cloud computing. *Wireless Personal Communications*, *73*(1), 51–61. https://doi.org/10.1007/s11277-012-0977-8
30. **Sha, K.***, Gehlot, J., & Greve, R. (2013). Multipath routing techniques in wireless sensor networks, a survey. *Wireless Personal Communications*, *70*(2), 807–829. https://doi.org/10.1007/s11277-012-0723-2
31. **Sha, K.**, Wang, S., & Shi, W. (2010). RD4: Role-differentiated cooperative deceptive data detection and filtering in VANETs. *IEEE Transactions on Vehicular Technology*, *59*(3), 1183–1190. https://doi.org/10.1109/TVT.2010.2040400
32. **Sha, K.**, & Shi, W. (2008). Consistency-driven data quality management in wireless sensor networks. *Journal of Parallel and Distributed Computing*, *68*(9), 1207–1221. https://doi.org/10.1016/j.jpdc.2008.06.004
33. Xi, Y., **Sha, K.**, Shi, W., Schwiebert, L., & Zhang, T. (2008). Probabilistic adaptive privacy-preserving authentication in vehicular networks. *Journal of Computer Science and Technology*, *23*(6), 916–928. https://doi.org/10.1007/s11390-008-9194-z
34. Du, J., Shi, W., & **Sha, K.** (2009). Asymmetry-aware link layer services in wireless sensor networks. *Journal of Embedded Computing*, *3*(2), 141–154. https://dl.acm.org/doi/10.1145/2069087.2069092
35. **Sha, K.**, Du, J., & Shi, W. (2006). WEAR: A balanced, fault-tolerant, energy-efficient routing protocol for wireless sensor networks. *International Journal of Sensor Networks*, *1*(1/2), 114–125. https://doi.org/10.1504/IJSNET.2006.012031
36. **Sha, K.**, & Shi, W. (2005). Modeling the lifetime of wireless sensor networks. *Sensor Letters*, *3*(2), 126–135.

Conferences and Workshops Publications

1. Anjum, S., Yang, X., Gerrity, T., Ding, J., **Sha, K.**, Zheng, W., Fu, S., Paes, A., and Feng, Y. (2026, October). Beyond Vibe Coding: Unified Multi-Framework Quantum Code Generation and Execution-Aware Evaluation with LLMs, In *Proceedings of 2026 IEEE 2nd International Conference on Quantum Artificial Intelligence (IEEE QAI 2026)*. Nottingham, UK.

2. Akhand, M. N. T., Arafat, A. A., **Sha, K.**, and Ahmed, M. R. (2026, October). MAST: A Two-Stage Mission-Aware Framework for Detecting Temporal Misalignment Attacks, In *Proceedings of the Eleventh ACM/IEEE Symposium on Edge Computing (IEEE/ACM SEC 2026)*. Santa Clara, CA, United States.
3. Cherukuri, K. S., Feng, Y., **Sha, K.*** (2026, October). Foundation Models for Edge Intelligence in Autonomous Driving: Opportunities and Challenges. In *Proceedings of the Eleventh ACM/IEEE Symposium on Edge Computing (IEEE/ACM SEC 2026)*. Santa Clara, CA, United States.
4. Chandrasekaran, A., Zikos, D., & **Sha, K.*** (2026, October) Beyond Prediction Confidence: Cross-Layer and Utility-Aware Uncertainty Quantification for Trustworthy Clinical AI, in *Proceedings of The 2026 IEEE International Conference on E-health Networking, Application & Services (IEEE Healthcom 2026),* New York City, NY, United States.
5. Gallardo, S., Pollyn, O., Meng, Z., **Sha, K** and Whitworth, C. (2026, October) Vulnerabilities of ROS 2 Communication in Autonomous Vehicle Systems, in *Proceedings of ACM/IEEE Joint Conference on Digital Libraries (ACM/IEEE JCDL) 2026,* Dallas, TX, United States. https://doi.org/10.1145/3805696.3847300
6. Wei, W., **Sha, K.**, & Chang, H.-C. (2026, August). Bridging AI governance and cybersecurity risk management: Topic modeling and framework-based gap analysis, in *Proceedings of the 2026 America's Conference on Information Systems (AMCIS 2026),* Reno, NV, United States. https://aisel.aisnet.org/amcis2026/ai_sigculture/ai_sigculture/5
7. Brishti, F., Mohammed, S., Chen, B., Wu, F., Feng, Y., & **Sha, K.** (2026, July). Enhanced DeepSMOTE for imbalanced classification with label noise, *in Proceedings of the 26th International Conference on Software Quality, Reliability, and Security (QRS 2026).* Florence, Italy.
8. Chandrasekaran, A., Zikos, D., Mete, M., Pang, A., Lund, B., & **Sha, K.*** (2026, August), NEURON: A Neuro-symbolic System for Grounded Clinical Explainability, in *Proceedings of the 11th IEEE/ACM conference on Connected Health: Applications, Systems and Engineering Technologies (ACM/IEEE CHASE 2026)*, Pittsburgh, PA, United States. https://doi.org/10.1109/CHASE69719.2026.00012 (**Best Paper Award**)
9. Wang, L., Juneja, S., Naar, S., Lee, M., **Sha, K.**, Shah, K., Green, S., Feng, Y. (2026, August) Generating High-Fidelity Synthetic Mental Health Records with Expert-in-the-Loop Verification, in *Proceedings of the 11th IEEE/ACM conference on Connected Health: Applications, Systems and Engineering Technologies (ACM/IEEE CHASE 2026)*, Pittsburgh, PA, United States. https://doi.org/10.1109/CHASE69719.2026.00023
10. Wang, L., Juneja, S., Naar, S., Lee, M., **Sha, K.**, Shah, K., Smith, A., Green, S., Feng, Y. (2026, August) LOGAR: A Local–Online Generate–Audit–Refine Framework for Mental Health De-identification, in *Proceedings of the 11th IEEE/ACM conference on Connected Health: Applications, Systems and Engineering Technologies (ACM/IEEE CHASE 2026)*, Pittsburgh, PA, United States. https:// doi.org/10.1109/CHASE69719.2026.00064
11. Wu, B. & **Sha, K.** (2026, July). Temporal Deviation Learning via FKAN-Enhanced Self-Supervision for Cyber-Physical Intrusion Detection, in *Proceedings of the 35th International Conference on Computer Communications and Networks (ICCCN 2026)*, Honolulu, HI, USA. https://doi.org/10.1109/ICCCN69946.2026.11662891
12. Feng, Y., Yang, X., **Sha, K.**, & Ding, J. (2026, April). Benchmarking the benchmarks for quantum code generation with LLMs. In *Proceedings of the 19th IEEE Dallas Circuits and Systems conference (DACS 2026)*, Dallas, TX, United States. https:// doi.org/10.1109/DCAS69364.2026.11545019
13. Zhou, Y., Sattari, M., Chen, H., & **Sha, K.*** (2026, May). Modeling and measuring redundancy in multisource multimodal data for autonomous driving. In *Proceedings of the 2026 IEEE International Conference on Mobility: Operations, Services, and Technologies (IEEE MOST 2026)*, Detroit, MI, United States. https://doi.org/10.1109/MOST69733.2026.00011 (**Best Paper Award**)
14. Chandrasekaran, A., Lund, B., Cherukuri, K. S., Mete, M., Zikos, D., and **Sha, K.** (2026, February) Integrating Narrative Explanations to Improve Clinical Sensemaking, in *2026 Multi-disciplinary Information Research Symposium (MIRS)* (**Best Paper Award, 1ST Place**)
15. Adarbah, H. Y., **Sha, K.**, & Noore, A. (2025, December). Toward design of a scalable federated unlearning framework for trustworthy edge intelligence. In *Proceedings of the Tenth ACM/IEEE*

Symposium on Edge Computing (SEC 2025). Washington, DC, United States. https://doi.org/10.1145/3769102.3774631

16. Cherukuri, K. S., **Sha, K.***, & Ding, J. (2025, August). Enabling federated learning for object detection in connected autonomous driving using YOLO with the Flower framework. In *Proceedings of the 34th International Conference on Computer Communications and Networks (ICCCN 20025)*, Tokyo, Japan. https://doi.org/10.1109/ICCCN65249.2025.11133941
17. Zhang, H., Fan, H., **Sha, K.**, Huang, Y., & Feng, Y. (2025, July). DAM: Dynamic attention mask for long-context large language model inference acceleration. In *Findings of the Association for Computational Linguistics: ACL 2025*. Association for Computational Linguistics. Vienna, Austria. https://doi.org/10.18653/v1/2025.findings-acl.242
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Journal Editorial

1. Sha, K., Liang, X., & Li, W. (2018). Editorial: Special issue on security in medical cyber-physical systems [Editorial]. *Smart Health*, *9-10*, 1–2.
2. Mao, S., Wang, H., Striegel, A., & Sha, K. (2016). Editorial: Special issue on cognitive, cellular, and mobile networks [Editorial]. *EAI Endorsed Transactions on Wireless Spectrum*, *16*(7), Article e1.
3. Parakh, A., & Sha, K. (2016). Editorial: Privacy, security and trust in mobile and wireless systems [Editorial]. *International Journal of Security and Networks*, *11*(1/2), 1–2.
4. Sha, K., & Wu, Z. (2013). Editorial: Privacy, security and trust in mobile systems [Editorial]. *Wireless Personal Communications*, *73*(1), 1–3.

Poster Papers

1. Niswanger, R., Sha, K., & Lerman, D. C. (2023, June).“Design of Mixed Reality Dangerous Situations for Autistic Children: Road Safety.” in *Proceedings of 2023 ACM/IEEE CHASE conference*, June 2023, Orlando, FL.
2. Wu, Y., Sha, K., & Yue, K.-B. (2022, December). “Blockchain-Enabled Federated Edge Learning for Big Data Quality Assessment.” in *Proceedings of 2022 IEEE/ACM SEC conference, Dec 2022, Seattle, WA*
3. Sun, H., Chen, Y., Sha, K., & Wu, Y. (2022, December). “Ensemble Federated Edge Learning for Recommender Systems.” in *Proceedings of 2022 IEEE/ACM SEC conference, Dec 2022, Seattle, WA*
4. Sha, K., Manghani, P., & Lerman, D. C. (2022, November). “Augmented Reality Game Therapy for Children with Autism Spectrum Disorder.” in *Proceedings of 2022 ACM/IEEE CHASE conference*, Nov 2022, Washington, DC.
5. Sha, K., & Li, Y. (2019, November**).** “Design of Chord Ring and SDN Based Edge Infrastructure Management System.” (3-page poster summary), in *Proceedings of The Fourth ACM/IEEE Symposium on Edge Computing*, Nov 2019, Washington, DC.
6. Sha, K., Liu, Z., & Dempsey, J. (2019, September**).** “Augmented Reality Based Therapy System for Social Skill Deficits.” (2-page poster summary), in *Proceedings of The Fourth IEEE/ACM Conference on Connected Health: Applications, Systems, and Engineering Technologies (IEEE/ACM CHASE 2019)*, Sep 2019, Washington, DC.
7. Kayastha, N., & Sha, K. (2019, September). “A Novel and Efficient Approach to Evaluate Biometric Features for User Identification.” (2-page poster summary), in *Proceedings of The Fourth IEEE/ACM Conference on Connected Health: Applications, Systems, and Engineering Technologies (IEEE/ACM CHASE 2019)*, Sep 2019, Washington, DC.
8. Sha, K., & Kumari, M. (2018, September). “Patient Identification Based On Wrist Activity Data.”, in *Proceedings of The Third IEEE/ACM Conference on Connected Health: Applications, Systems, and Engineering Technologies (IEEE/ACM CHASE 2018)*, Sep 2018, Washington, DC.
9. Manoharan, N., Sha, K., & Lu, J. (2017, July). “Exercise Evaluation Using Wrist Sensors.” in *Proceedings of The Second IEEE/ACM Conference on Connected Health: Applications, Systems, and Engineering Technologies (IEEE/ACM CHASE 2017)*, July 2017, Philadelphia, PA.
10. Kelagote, N., Sha, K., Lu, J., Sun, L., & Kadiwal, S. (2016, October). “Human Motion Modeling and Evaluation Using Wearable Sensors.” in *Proceedings of the Dual Conference 2016 in Innovation and Automation*, October 2016, Houston, USA.
11. Kadiwal, S., Lu, J., Sha, K., Sun, L., & Kelagote, N. (2016, October) “Microsoft Kinect Based Real-time Segmentation and Recognition for Human Activity Learning.” in *Proceedings of the Dual Conference 2016 in Innovation and Automation*, October 2016, Houston, USA.
12. Wang, S., Sha, K., & Shi, W. (2009, April). “Role-based Deceptive Data Detection and Filtering in WSNs.” in *Proceedings of the 8th ACM/IEEE International Conference on Information Processing in Sensor Networks (IPSN 2009)* (2 page summary), April 13-16, 2009, San/Francisco.
13. Sha, K., & Shi, W. (2004, November). “Revisiting the Lifetime of Wireless Sensor Networks.” in *Proceedings of the 2nd International Conference on Embedded Networked Sensor Systems (SenSys’04)* (poster), November 3-5, 2004

Poster Presentation

1. Montazeri, S., Sha, K., Dargie, W., & Feng, Y. (2026, November). “Activity Recognition based on ECG only”, in *UHT HSC RAD 2026*, April 2026, Dallas, TX.
2. Cherukuri, K. S., & Sha, K. (2026, April). “Evaluating Federated Learning for Autonomous Driving”, in *UNT InfoForce*, April 2026, Frisco, TX **(1st place winner)**
3. Chung, H., Ding, J., Sha, K. (2026, April). *Turning Inclusion into Evidence: Data-Informed Mentoring for Student Well-being and Success*, in *UNT InfoForce*, April 2026, Denton, TX
4. Montazeri, S., Sha, K., Dargie, W., & Feng, Y. (2025, November). “Activity Recognition based on ECG only”, in *2025 One Health Collaborative Symposium*, November 2025, Dallas, TX.
5. Cherukuri, K. S., & Sha, K. (2025, October). “Evaluating Federated Learning for Autonomous Driving”, in *UNT AI & Data Science Executive Summit*, October 2025, Frisco, TX

6. Zhou, Y., Chen, H., & Sha, K. (2025, October). "A Novel Multi-layer Task-centric and Data Quality Framework for Autonomous Driving.", in *UNT AI & Data Science Executive Summit*, October 2025, Frisco, TX
7. Zhang, H., Fan, H., Sha, K., Huang, Y., & Feng, Y. (2025, October). "DAM: Dynamic Attention Mask for Long-Context Large Language Model Inference Acceleration." in *UNT Research Day*, October 2025, Denton, TX
8. Cherukuri, K. S., & Sha, K. (2025, October). "Evaluating Federated Learning for Autonomous Driving", in *UNT Research Day*, October 2025, Denton, TX
9. Montazeri, S., Sha, K., Dargie, W., & Feng, Y. (2025, September). "Activity Recognition based on ECG only", in *2025 Texas Health Informatics Alliance Conference*, September 2025, Arlington, TX.
10. Zhou, Y., Tu, F., Sha, K., Ding, J., & Chen, H. (2024, September). "A Survey on Data Quality Dimensions and Tools for Machine Learning", in *Day of Health Informatics and Data Science*, September 2024, Frisco, TX. (the **3rd prize** for Data Science student poster competition)
11. Cherukuri, K. S., Sha, K., & Chen, H. (2024, September). "Evaluation of Privacy-Preserving Techniques for Edge Computing", in *Day of Health Informatics and Data Science*, September 2024, Frisco, TX. (the **2nd prize** for Data Science student poster competition).
12. Sha, K. "EdgeSec: Edge Computing for IoT Security", in *Proceedings of Sustainable Research Pathways Workshop* organized by SHI Sustainable Research Pathways Institute 2017, Dec 2017, Lawrence Berkeley National Laboratory, CA.
13. Reddy, R., Sha, K., Gehlot, J., & Greve, R. (2011, November). "On the Heterogeneity in Mobile Application Development." in *Proceedings of Oklahoma Research Day*, Nov 2011, Lawton, Oklahoma, USA

Paper Under Submission/In Preparation

1. Cherukuri, K. S., Whitworth, C., Huang, Z. & **Sha, K.*** (2025). *Enabling federated object detection for connected autonomous vehicles: A deployment-oriented evaluation* (arXiv:2509.01868) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2509.01868 [Manuscript submitted for publication]
2. Cherukuri, K. S., Shi, W., Ding, J., & **Sha, K.*** (2026). *Deployment oriented federated learning in connected and autonomous vehicles: Challenges and opportunities* (Technical Report) [Manuscript submitted for publication].
3. Cherukuri, K. S., & **Sha, K.*** (2026). *An Adaptive LLM-Assisted Framework for Training Scheduling in Multi-Model Federated Learning for Autonomous Vehicles* (Technical Report) [Manuscript submitted for publication].
4. Dargie, W., Hegde, S. S., & **Sha, K.** (2025). *Person identification using footstep acoustics with CNN and Transformer models in real-world settings* [Preprint]. SSRN. http://dx.doi.org/10.2139/ssrn.5232180
5. Wang, L., **Sha, K.**, Feng, Y., Juneja, S., et al. (2025). *Systematic review of the capabilities of LLM-based tools in mental health and proposal of a clinically informed evaluation framework* (Technical Report) [Manuscript submitted for publication].
6. Wang, L., Madison, L., Junzhang, J., **Sha, K.**, et al. (2026). *Initial risk probing and feasibility testing of Glow: A generative AI-powered dialectical behavior therapy skills coach for substance use recovery and HIV prevention* (arXiv:2602.08121) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2602.08121
7. Wang, L., Madison, L., Chen, S., **Sha, K.**, Feng, Y., Hightow-Weidman, L., Yuwen, W. (2026). *Human-Centered Evaluation of a GenAI-Powered Dialectical Behavior Therapy Skills Coach for Substance Use Recovery: Acceptability, Feasibility, Usability, and Safety*. (Technical Report) [Manuscript submitted for publication].
8. Wang, L., Madison, L., Chen, S., **Sha, K.**, Feng, Y., Hightow-Weidman, L., Yuwen, W. (2026). *Initial Risk Probing of Glow: a Generative AI-Powered Dialectical Behavior Therapy Skills Coach for Substance Use Recovery and HIV Prevention*. (Technical Report) [Manuscript submitted for publication].

9. Zhang, H., Zhang, Q., Cao, Q., Fan, H., Huang, Y., **Sha, K.**, & Feng, Y. (2026). *TileQuant: Tile-Level Mixed-Precision Post-Training Quantization for Large Language Models* (Technical Report). [Manuscript submitted for publication]
10. Zhang, H., Zhang, Q., Cao, Q., Fan, H., Huang, Y., **Sha, K.**, & Feng, Y. (2026). *TileMix: Tile-Centric Mixed-Precision Attention for LLM Inference Acceleration* (Technical Report). [Manuscript submitted for publication]
11. Tang, B., **Sha, K.**, Arafin, T., Allison, M., et al. (2026). *On the vulnerabilities of Ant-Q multi-agent reinforcement learning algorithm* (Technical Report). [Manuscript submitted for publication]
12. Gerrity, T., Fu, S., Yang, X., Ding, J., **Sha, K.**, Zheng, W., and Feng Y. (2026). *Quantum Bloom Filters for Privacy-Preserving Medical Data Protection and Analytics* (Technical Report) [Manuscript submitted for publication]
13. Anjum, S., Yang, X., Ding, J., **Sha, K.**, Zheng, W., Fu, S., and Feng Y. (2026). *Beyond Vibe Coding: Unified Multi-Framework Quantum Code Generation and Execution-Aware Evaluation with LLMs* (Technical Report) [Manuscript submitted for publication]
14. Dong, S., Meng, Z., Sun, H., **Sha, K.**, Fan, H. (2026). *RGBD20K: A Large-Scale Benchmark for RGB-D Semantic Segmentation* (Technical Report) [Manuscript submitted for publication]
15. Li, C., Qi, X., Zhang, Y., Yuan, X., Wu, K., Ding, J., **Sha, K**., Liu, R., and Ni, W., (2026) *QUAC: Verified Bi-Level Quantum-Assisted PID Control for Differentially Private Federated Learning* (Technical Report) [Manuscript submitted for publication]
16. Meng, Z.*, Liang, J.*, Jagathpally, A.*, Feng, Y., Sha, K., Fan, H. (2026). *VTRVOS: A Visible-Thermal Benchmark for Referring Video Object Segmentation* (Technical Report) [Manuscript submitted for publication]

Supervised Master Thesis

1. Trada, P. *"Evaluating Sentiment Analysis Mechanisms For Amazon Review"*, May 2023
2. Kadve, R. *"Federated Learning to Build Sentiment Analysis Models for Amazon Review Dataset Without Labels"*, May 2023
3. Kumari, P *"Tagging Negative Review in Amazon Product Review"*, May 2023
4. Sudhakar, S. R. V. *"An Efficient Classification Algorithm for Gait Authentication based on Streaming Sensory Data"*, May 2021
5. Kayastha, N. "Biometrics-based User Identification with Optimal Feature Evaluation and Selection", mater thesis, Dec 2019
6. Ramadhan, M. *"EBMR: Energy-Efficient and Load-Balanced Multipath Routing Protocol for Wireless Sensor Networks"*, master thesis, May 2016
7. Basheer, A *"QAAC: Quality-Assured Adaptive Data Compression for Sensor Data"*, master thesis, Dec 2015

Recent Talks

Invited talks or keynote talks

- *Beyond Models: A Data Quality-Centric Approach to Autonomous Driving,* invited talk at AI Colloquium, South Dakota State University, September 2026.
- *From Data to Decisions,* invited talk at International Seminar: STIKOM Uyelindo Kupang-NTT, Indonesia, June 2025.
- *Towards Efficient, Secure, and Quality-aware Data Analytics*, **Keynote** talk at the 4th Interdisciplinary Conference on Electrics and Computer, Chicago, IL, June 2024
- *Edge-computing Supported Robust Multi-Robot Systems*, talk at Information Science Brown Bag Research Meeting at University of North Texas, Sep 2023
- *User Identification, Authentication, and Secure Storage*, Invited talk at University of Houston, March 2023

- *ActID: Activity Sensor Based User Identification*, Invited virtual talk at Florida Gulf Coast University, Jan 2023
- *Efficient, Secure, and Privacy-preserving Data Analytics and Sharing*, Invited talk at University of North Texas, Dec 2022
- *How to write research papers*, invited virtual talk at Anhui University, November, 2021
- *Cybersecurity Research at UHCL*, invited talk at UHCL, October, 2021
- *ActID: An Efficient Framework for Activity Sensor Based User Identification*, Invited virtual talk at Nanjing University of Posts and Communications, May 2021
- *Data Quality Management in IoT,* invited virtual talk at Anhui University, November, 2020
- *Edge Computing Based Designs for Security and Data Analytics,* invited talk at North Carolina Central University, April 28, 2020
- *Edge Computing Based Designs for Security, Performance Optimization, and Data Analytics,* invited talk at Oakland University, January 23, 2020
- *Edge Computing based Security Designs*, invited talk at SIG of InfraGard Defense Industrial Base, Jan 14, 2020
- *Studies on Edge Computing for IoT Security,* invited seminar talk at Wayne State University, March 27, 2019
- *Research on Edge Computing Security,* invited talk at Anhui University, August 2018
- *Home-based Exercise Evaluation Using Wrist Sensors*, Keynote at Healthcare IT 2018 at World Congress & Expo on Healthcare IT, August 2018
- *Security Issues in Edge Computing,* Colloquium talk at University of Houston – Clear Lake, April 23, 2018
- *Introduction to Internet of Things,* Invited talk at Math Center at University of Houston – Clear Lake, September 2017
- *Quality-Assured Adaptive Data Compression for Sensor Data,* Invited talk at Nanjing University of Posts and Communications, June 2017
- *Emerging Cyber Threat: Challenges & Opportunities*, Keynote at University of Houston System Cybersecurity Cyber-Resilience Workshop, March 2017
- *IoT Security: challenges and solutions*, invited talk at Zhejiang Wanli University, January 2017
- *Security Issues in Internet of Things: opportunities and challenges*, invited talk at Ningbo University, January 2017
- *IoT Security: opportunities and solutions*, invited talk at Shanghai University of International Business and Economics, January 2017
- *Security in Internet of Things: vision and challenges*, Invited talk at Anhui University, December 2016
- *Edge Supported Security Services for Internet of Things*, Invited talk at Nanjing University of Posts and Communications, December 2016
- *IoT Security: vision, challenges, and solutions*, Keynote at International Conference on Wireless Communication and Sensor Network (WCSN 2016), December 2016
- *IoT Security: Perspectives, Challenges, and Cases*, invited Round Table talk at University of Nebraska at Omaha, April 2016
- *Energy Efficient Data Quality Management in Wireless Sensor Systems*, invited talk at University of Houston – Clear Lake, May 2015
- *A Secure Framework to Read Isolated Devices in Smart Grid*, invited talk at Howard University, Apr 2015
- *Data Quality Management in WSN*, invited talk at Eastern Washington University, Apr 2015
- *Role-based Deceptive Data Detection in VANET,* invited talk at University of Indianapolis, Feb 2015
- *A Cloud-aided Framework for Securely Reading Isolated Devices in Smart Grid*, invited talk at Nanjing University of Posts and Communications, July 2014
- *An Efficient Framework for Data Quality Management*, invited talk at Nanjing University of Posts and Communications, July 2014

- *Data Quality in Wireless Sensor Networks*, invited talk at Oklahoma City University, Oct 2013

Conference presentations

- *Modeling and measuring redundancy in multisource multimodal data for autonomous driving.* the 2026 IEEE International Conference on Mobility: Operations, Services, and Technologies (MOST 2026), Detroit, MI, May 2026
- *Toward Design of a Scalable Federated Unlearning Framework for Trustworthy Edge Intelligence*, the Tenth ACM/IEEE Symposium on Edge Computing (ACM/IEEE SEC 2025), Washington DC, December 2025
- *Enabling Federated Learning for Object Detection in Connected Autonomous Driving Using YOLO with the Flower Framework,* 34th International Conference on Computer Communications and Networks (ICCCN 2025), Tokyo, Japan, August 2025.
- *D2SO: Detecting Distant and Small Objects for Vision-based Vehicle Autonomous Systems.* 2025 IEEE International Conference on Mobility: Operations, Services, and Technologies (MOST 2025), Newark, DE, May 2025
- *Big Data Quality Scoring for Structured Data Using MapReduce,* 33rd International Conference on Computer Communications and Networks (ICCCN 2024), Big Island, HI, July 2024
- *An Edge-Based Approach to Robust Multi-Robot Systems in Dynamic Environments,* poster presentation, 2024 CISE Research Expansion Grantee Meeting, Denver, CO, May 2024
- *Effective Collaborations among Six Institutions,* 2024 CISE Research Expansion Grantee Meeting, Denver, CO, May 2024
- *Edge Computing Designs for IoT Security,* talk at HSI Summer Camp, July 15, 2022
- *An Introduction to NICE Cybersecurity Workforce Framework*, talk at Cybersecurity Education Workshop, April 5, 2019
- *The Design of Network Forensics Course*, talk at Cybersecurity Education Workshop, April 5, 2019
- *The World of IoT*, invited talk at The Chinese Entrepreneur Association, April 2016

Panel Moderator

- *Industry Panel: Industry-Academia Collaboration in Healthcare*, the 11th IEEE/ACM conference on Connected Health: Applications, Systems and Engineering Technologies, August, 2026.

Panelist

- *NSF proposal review procedures and tips*, 2025-2026 NSF CISE MSI Aspiring PI Virtual Workshop, January, 2025-2026.
- *Preparing Budget for NSF proposals,* 2025-2026 NSF CISE MSI Aspiring PI Virtual Workshop, January 2025-2026
- *The Impact of AI of the Current and Next Generation of data Practitioners*, The Consero Chief Data & AI Officer Executive Summit, Dallas, November, 2025.
- *Preparing Competitive NSF Proposals of Engineering and Computing Education,* NSF Proposal Writing Workshop for Hispanic-Serving Institution, 2023 IEEE ASEE Frontiers in Education Conference, College Station, Texas, October 18, 2023
- *Security Education and Research at UNT,* Cyber Awareness Month Series – Panel Discussion, University of North Texas, September 15, 2023
- *Doing Research at UHCL,* Faculty Development Day at University of Houston – Clear Lake, April 6, 2018
  Security Education and Research at UHCL, University of Houston System Cybersecurity Cyber-Resilience Workshop, March 10, 2017

Professional and Scholarly Participation

Funding Proposal Reviewer

- NSF Panelist, 2026
- NSF Panelist, 2025
- Hong Kong Research Grants Council, 2024, 2026
- Oakland University Research Grant, 2024
- NSF Panelist, 2021
- NSF Panelist, 2020
- NSF Panelist, 2017
- Sam Houston State University Research Fund, 2017
- NSF Bluewaters Scholarship, 2014, 2015, 2016, 2017, 2018
- University of Central Oklahoma On-campus Interdisciplinary Grant, 2014
- University of Pisa Research Proposal 2008

Professional Membership

- Senior Member of IEEE
- Senior Member of ACM

Professional Services

- **Editorship**
  - **Associate Editor-in-Chief (EiC)**
    - Smart Health (Elsevier), 2026 -
  - **Book Series Editor,** Sustainable Industrial Engineering and Management Systems book series, CRC Press, Routledge Taylor & Francis Group
  - **Associate Editor & Editorial Board Member**
    - IEEE Journal of Biomedical and Health Informatics, 2026 -
    - IEEE IoT Journal, 2021 -
    - Smart Health (Elsevier), 2019 -
    - Computing (Springer), 2020 -
    - Multimedia Tools and Applications (Springer), 2023 –
    - Future Internet, 2026 -
  - **Guest Editor**
    - Special Issue: Trustworthy, Secure, and Intelligent Healthcare Systems: Advances in AI and Connected Health, *Future Internet*, 2026-2027
    - Special Issue: Advances in Generative AI and Computational Linguistics, *Electronics*, 2025-2026
    - Special Issue in Advanced Deep Learning and Sensing Techniques for Complex Structural Health Monitoring, *Sensors*, 2022-2023
    - Special Issue in New Advances in Securing Data and Big Data, *Algorithms*, 2021-2022
    - Special Issue in Security and Privacy in Internet of Medical Things (IoMT), *Security and Communication Networks*, 2019-2020
    - Topic collections in Security and Privacy in Smart Connected Health Systems, *Journal of Medical Systems*, 2018-2019
    - Special Issue in Security of Medical CPS, *Elsevier Smart Health*, 2017-2018
    - Special Issue in Privacy, Security and Trust in Mobile Systems, *Journal of Security and Network*, 2016
    - Special Issue in *EAI Endorsed Transactions on Wireless Spectrum*, 2016

        - Special Issue in Mobile Privacy, Security and Trust, *Personal Wireless Communications*, 2013

- **Organization Officer/Secretary**
    - Founding Board of Directors and Secretary of International Autonomous Driving Organization, 2021 - present
    - Executive Committee Member and Communication Director, Association of Chinese Scholars in Computing, 2023 - present
    - Secretary of Technical Committee on the Internet of the IEEE Computer Society (IEEE-CS TCI) (2013 - 2016)
    - Founding Member, IEEE Internet of Wearable Things (IoWT) Working Group, 2025 -

- **Conference Organizing Committee & Chair**
    - **Steering Committee Member/Organizing Committee Advisory Member**
        - IEEE International Conference on Mobility: Operations, Services, and Technologies (IEEE MOST), 2025 -
        - International Conference on Computer Communications and Networks (IEEE ICCCN), 2023-
    - **Conference General Chair**
        - The 2nd IEEE Mobility: Operations, Services, and Technologies (MOST 2024)
        - The 8th ACM/IEEE Symposium on Edge Computing (SEC 2023)
        - The 6th IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technology (CHASE 2021)
        - The Third International Conference on Wireless Communication and Sensor Network (WCSN 2016)
    - **Program co-Chair**
        - NSF CISE Research Expansion (RE) Aspiring PI Workshop at FIU, 2025-2026
        - The 3rd International Conference on Cybersecurity and Artificial Intelligence Strategies (CAIS'26)
        - NSF CISE MSI Aspiring PI Workshop at UNT, 2024-2025
        - The 2nd International Conference on Cybersecurity and Artificial Intelligence Strategies (CAIS'25)
        - The 24th International Conference on Computer Communications and Networks (ICCCN 2015)
    - **Workshop General Chair**
        - The International Conference on Computer Communications and Networks (ICCCN 2018-2027)
        - The 11th IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technology (CHASE 2026)
        - The 2026 IEEE International Conference on E-health Networking, Application & Services (IEEE Healthcom 2026)
        - The 10th IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technology (CHASE 2025)
        - The 22nd International Conference on Computer Communications and Networks (ICCCN 2013)
        - The 9th ACM/IEEE Symposium on Edge Computing (SEC 2024)
    - **Symposium Chair**
        - The 2019 International Conference on Computing, Networking and Communications (ICNC 2019)
    - **Track Chair**
        - The 26th International Conference on Software Quality, Reliability, and Security (QRS 2026)
        - The 26th International Conference on Computer Communications and Networks (IEEE ICCCN 2017)
    - **Student Travel Grant Chair**

- The 14th IEEE International Conference on Sensing, Communication and Networking (IEEE SECON 2017)
- The 2nd IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technology (CHASE 2017)
- The 25th International Conference on Computer Communications and Networks (ICCCN 2016)

- **Publication Chair**
  - The First IEEE International Conference on Mobility: Operations, Services, and Technologies (MOST 2023)
  - The ACM/IEEE Symposium on Edge Computing (SEC 2020-2022)
  - The IEEE International Conference on Connected and Autonomous Driving (MetroCAD 2020-2022)
  - The IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technology (2017-2020)
- **Workshop Co-Chair**
  - The Nineth ACM/IEEE Workshop on Security and Privacy in Edge Computing (EdgeSP 2026)
  - The Eighth ACM/IEEE Workshop on Security and Privacy in Edge Computing (EdgeSP 2025)
  - The Workshop on Synergy of AI and Multimodal Biomedical Data Mining (co-located with ICDM 2025)
  - The International Workshop on Security, Privacy, and Trustworthiness in Medical Cyber Physical Systems (MedSPT 2016-2018)
  - University of Houston System Cybersecurity Cyber-Resilience Workshop, 2017
  - The International Workshop on Privacy, Security and Trust in Mobile and Wireless Systems (MobiPST 2011, 2012, 2014)
- **Session Chair**
  - The 11th IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technology (CHASE 2026)
  - The International Conference on Computer Communications and Networks (ICCCN 2011, 2013-2017, 2020, 2026)
  - The 4th IEEE International Conference on Mobility: Operations, Services, and Technologies (MOST 2026)
  - The 10th IEEE Symposium on Edge Computing (SEC 2025)
  - The 2nd IEEE International Conference on Mobility: Operations, Services, and Technologies (MOST 2024)
  - The 8th IEEE Symposium on Edge Computing (SEC 2023)
  - The 1st IEEE International Conference on Mobility: Operations, Services, and Technologies (MOST 2023)
  - The 4th IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technology (CHASE 2019)
  - The 17th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGrid 2017)
  - The 3rd International Workshop on Context-aware QoS Provisioning and Management for Emerging Networks, Applications and Services (ContextQoS 2013)
  - The International Conference on Collaborative Computing (CollaborateCom 2008-2009)
- **Publicity Chair**
  - International Conference on Computing, Networking and Communications (ICNC 2017)
  - The 2017 International Conference on Applications and Techniques in Cyber Intelligence (ATCI 2017)
  - 2013 International Workshop on Mobile Cloud Computing Technologies and Applications (MCCTA 2013)

        - The 3rd International Workshop on Privacy, Security and Trust in Mobile and Wireless Systems (MobiPST 2013)

- **TPC member**
    - The Eleventh ACM/IEEE Symposium on Edge Computing (ACM/IEEE SEC 2026)
    - 2026 IEEE Vehicular Networking Conference (IEEE VNC 2026)
    - The Fourth IEEE International Conference on Mobility: Operations, Services, and Technologies (IEEE MOST 2026)
    - The First IEEE Workshop on Physical Intelligence: Systems and Applications (IEEE PISA 2026)
    - The IEEE International Conference on Trust, Security and Privacy in Computing and Communications (IEEE TrustCom 2025)
    - 2024 IEEE World Forum on Public Safety Technology (IEEE WF-PST)
    - IEEE International Conference On Artificial Intelligence Testing (AITesting 2024)
    - The IEEE International Conference on Communications (IEEE ICC 2016，2019-2021, 2023-2026)
    - International Conference on Computing, Networking and Communications (ICNC 2023-2024)
    - The 8th IEEE International Conference on Intelligent Data and Security (IEEE IDS 2022)
    - The 15th International Conference on Knowledge Science, Engineering and Management (KSEM 2022)
    - The IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technology (CHASE 2018-2021)
    - The IEEE International Conference on Communications (IEEE ICC 2016，2019-2021, 2023-2024)
    - 2021 4th International Conference on Computer Applications & Information Security (ICCAIS'2021)
    - The International Conference on Computer Communications and Networks (ICCCN 2013，2015-2017，2020-2021)
    - International Conference on Algorithms and Architectures for Parallel Processing (ICA3PP 2020)
    - The International Conference on Wireless Algorithms, Systems, and Applications (WASA 2016-2021)
    - The 2020 International Conference on Computing, Networking and Communications (ICNC 2020)
    - The 25th IEEE International Conference on Parallel and Distributed Systems (ICPADS 2019)
    - The 4th IEEE International Conference on Intelligent Data and Security (IEEE IDS 2019)
    - The IEEE International Conference on Computational Science and Engineering (IEEE CSE 2012-2013，2019)
    - The International Conference on Green, Pervasive and Cloud Computing (GPC 2013，2015，2017-2019)
    - National Cyber Summit (NCS 2018-2021)
    - The 2nd International Workshop on Attacks and Defenses for Internet-of-Things (ADIoT 2019)
    - The IEEE International Conference on Ubiquitous Computing and Communications (IEEE IUCC 2017-2018)

- The IEEE International Conference on Trust, Security and Privacy in Computing and Communications (IEEE TrustCom 2016-2018)
- The 2017 IEEE International Conference on Collaboration and Internet Computing (IEEE CIC 2015-2017)
- The IEEE Conference on Local Computer Networks (IEEE LCN 2016-2017)
- International Conference on Agents and Artificial Intelligence 2017
- The International Workshop on Cloud Storage Service and Computing, 2016-2017
- 2017 IEEE WCNC Workshop on M2M Communications and the Internet of Things
- The 35th IEEE International Performance, Computing, and Communication Conference (IPCCC 2016)
- The annual International Conference on Distributed Computing in Sensor Systems (DCOSS 2016)
- The International Conference on Security of Smart Cities, Industrial Control System and Communications (SSIC 2016)
- The 2015 Global Conference on Information Technology, Computing, and Applications (Global IT 2015)
- The Third International Symposium on Security in Computing and Communications (SSCC'15)
- The Sixth International Conference on Mobile Computing, Applications and Services (MobiCASE 2014)
- The 5th International Conference on Creative Converged IT (CCIT 2014)
- The IEEE International Conference on High Performance Computing and Communications (IEEE HPCC 2013-2014)
- The 2014 FTRA International Symposium on Ubiquitous Green IT Systems (U-GIT 2014)
- The 2014 IEEE International Conference on Systems, Man, and Cybernetics (IEEE SMC 2014)
- The IEEE International Conference on Collaborative Computing: Networking, Applications and Worksharing (CollaborateCom 2009-2014)
- The Workshop on Big Data and e-Health, 2014
- The International Conference on Ambient Systems, Networks and Technologies (ANT), 2013
- The International Workshop on Internet of Ubiquitous and Pervasive Things (IUPT 2011-2014)
- The International Workshop on Data Management for Cyber Physical System (DM-CPS), 2013
- The World Conference on Information Systems and Technologies (WorldCIST) 2013
- The International Conference on Advanced Collaborative Networks, Systems and Applications (COLLA 2011-2014)
- The International Conference on Advanced Engineering Computing and Applications in Sciences (ADVCOMP 2011-2014)
- The International Conference on Mobile Web Information Systems (MobiWIS) 2012
- The International Workshop on Sensor Networks (SN) 2012
- The IEEE International Conference on Scalable Computing and Communications (IEEE ScalCom) 2012
- The IEEE International Conference on Wireless Communications, Networking and Information Security (IEEE WCNIS) 2010
- The IEEE International Workshop on Mobile Cyber-Physical Systems (MobiCPS) 2010
- The IEEE International Conference on Green Computing and Communications (IEEE GreenCom) 2010
- The International Workshop on Wireless Ad hoc and Sensor Networking (WWASN 2011)

- **Journal Reviewer**
  - IEEE Internet of Things Journal, 2016- 2022, 2024 - 2026

- ACM Transactions on Computing for Healthcare, 2025 - 2026
- ACM Transactions on Internet of Things, 2021-2022, 2024 - 2026
- ACM Computing Survey, 2024, 2026
- Nature Scientific Reports, 2025
- IEEE Transactions on Big Data, 2025
- Elsevier Smart Health, 2017-2024
- IEEE Transactions on Emerging Topics in Computing, 2023
- IEEE Transactions on Vehicular Technology, 2016-2022
- IEEE Transactions on Green Communications and Networking, 2021
- IEEE Sensor Journal, 2020
- IEEE Transactions on Smart Grid, 2020
- IEEE Journal of Selected Areas in Communications, 2020
- IEEE Transactions on Information Forensics & Security, 2020
- ACM Transactions on Computing for Healthcare, 2019-2021
- IEEE Internet Computing, 2019, 2020
- IEEE Transactions on Big Data, 2017, 2018, 2020
- IEEE Transactions on Industrial Informatics, 2018, 2019
- IEEE Transactions on Service Computing, 2017- 2019, 2021, 2023
- IEEE Transactions on Network Science and Engineering, 2018, 2020
- IEEE Transactions on Dependable and Secure Computing, 2014, 2020
- IEEE Network Magazine, 2020-2023
- IEEE Journal of Biomedical and Health Informatics, 2023
- Multimedia Tools and Applications, 2023
- International Journal of Machine Learning and Cybernetics, 2020
- Digital Communications and Networks, 2020
- Wireless Communications and Mobile Computing, 2020
- Computers, Materials & Continua, 2020
- Information Systems Frontiers, 2020
- ACM Transactions on Information Technology (TOIT), 2016-2017, 2019-2021
- IEEE ACCESS, 2017-2019
- Transactions on Emerging Telecommunications Technologies, 2019
- IEEE Transactions on Engineering Management, 2018
- IEEE Communications Letters, 2018
- IEEE Transactions on Mobile Computing, 2017, 2018
- IEEE/ACM Transactions on Networks, 2017, 2018
- IEEE Transactions on Parallel and Distributed Systems (TPDS) 2008, 2010-2017
- IEEE Transactions on Cloud Computing, 2017
- IEEE System Journal, 2015, 2018
- Elsevier Smart Health, 2017-2022
- Future Generation Computer Systems, 2018
- Journal of Network and Computer Applications, 2018
- IEEE Reviews in Biomedical Engineering, 2018
- Journal of Parallel and Distributed Computing (JPDC) 2007, 2010, 2011, 2018
- Sustainable Computing, Informatics and Systems, 2017
- EURASIP Journal on Wireless Communications and Networking, 2017, 2018
- IEEE Sensor Journal, 2016
- IEEE Internet Journal, 2016
- Computing (COMP), 2013, 2017, 2018
- International Journal of Communication Systems, 2017
- Security and Communication Networks, 2017, 2022
- Computational Intelligence, 2016
- Sensors, 2016, 2017
- Elsevie's Computer Networks (ComNet) 2011, 2014, 2016, 2017, 2018
- Elsevier's Computer Communications Journal (ComCom) 2008. 2011, 2012, 2013, 2014, 2016, 2020
- Wireless Networks, 2016

- Ad Hoc Networks 2013, 2014, 2015, 2016
- Future Internet, 2022
- IEEE Transactions on Computers, 2013
- IEEE Transactions on Intelligent Transportation Systems (TITS), 2013, 2023
- Transactions on Emerging Telecommunications Technologies, 2024
- International Journal of Security and Networks, 2015, 2016
- International Journal of Ad Hoc and Ubiquitous Computing 2009, 2010, 2013, 2016
- Wireless Personal Communications (WIRE) 2009-2014,2016
- IET Wireless Sensor Systems, 2015
- ETRI Journal 2014
- Journal of Systems Architecture, 2012, 2013
- Journal of Network and Computer Applications 2013, 2014
- International Journal of Sensor Networks (IJSNet) 2009, 2012-2014, 2016-2017
- Journal of Sensors 2009, 2012, 2014, 2015
- Mobile Networks and Applications (MONE), 2010, 2011, 2013
- International Journal On Advances in Intelligent Systems 2012-2014
- International Journal On Advances in Software 2012-2014
- International Journal of Parallel, Emergent and Distributed Systems, 2012
- IST 2012
- BMC Bioinformatics, 2012
- International Journal of Systems Science, 2010
- International Journal of Distributed Sensor Networks (IJDSN), 2011
- Journal of Computer Networks and Communications (JCNC), 2011
- Mathematical and Computer Modeling (MCM), 2010, 2011
- International Journal of Internet Protocol Technology (IJIPT) 2010
- International Journal of Intelligent Engineering & Systems 2009
- ACM Transactions on Autonomous Adaptive Systems (ACM TAAS) 2008
- Journal of Computer Science and Technology (JCST) 2008, 2010
- Journal of Computers and Applications 2007

- **Other Services**
  - Co-Organizer, Data Science Talk Series, 2024, 2025, 2026
  - Data Science Summer Camp Organizer, 2025, 2026
  - Founding Member, IEEE Internet of Wearable Things (IoWT) Working Group, 2025 -
  - UNT Data Science New Student Orientation, 2024
  - Mentor, NSF CISE MSI Proposal Writing, 2024, 2025, 2026
  - Ph.D. dissertation committee member, Information Science, UNT, 2024 – 2025
  - Ph.D. dissertation committee member, Computer Science and Engineering, UNT, 2024 -
  - Judge, UNT Data Science Poster Competition, 2024
  - Judge, 2024 UNT Multi-disciplinary Information Research Symposium (MIRS), 2024
  - Judge, UHCL STEM Challenge, 2022
  - Ph.D. dissertation committee member, Wayne State University, 2022-
  - Sponsor, Clear Lake High School STEM Challenger Team, 2022
  - Mentor, programming training for high school students, 2020
  - Board Member of Bay Area Youth Symphony, 2021
  - Mentor of High School Independent Study in IoT, 2021-2022
  - Invited External Reviewer of faculty promotion and tenure, University of Washington Bothell
  - Invited External Reviewer of faculty promotion and tenure, Saint Joseph's University
  - Invited External Reviewer of faculty promotion and tenure, University of Nebraska–Omaha
  - Invited External Reviewer of faculty promotion and tenure, Lewis University
  - Invited External Reviewer of faculty promotion and tenure, University of Massachusetts Boston
  - Judge of InnoSTARS Competition, 2018
  - Judge of FIRST Robotics Competition, Oklahoma Region, 2013, 2014

University Services

New Program Development

- *Ph.D. in Data Science,* University of North Texas
- *M.S. in Data Science,* University of North Texas (fully revision)
- *M.S. in Data Science,* University of Houston-Clear Lake
- *Minor in Cybersecurity for Computer Science Students,* University of Houston-Clear Lake

University Systems, University, College and Department Service

- Associate Chair, Department of Data Science (Fall 2026 -)
- Director, Data Science PhD Program (Fall 2024 –)
- Director, Data Science Master Program (Fall 2024 –)
- Director, Data Science Undergraduate Program (Spring 2025 – Summer 2025)
- Member, University Fellowship Review Committee, (Spring 2026)
- Member, UNT AI Governance Committee, (Fall 2026 -)
- Chair, Data Science Faculty Search Committee (Spring 2026)
- Chair, Data Science Curriculum Committee (Spring 2025)
- Chair, Data Science Personnel Affairs Committee (Spring 2025 – Fall 2025)
- Chair, Data Science Reappointment, Promotion and Tenure Committee (Spring 2025 – Fall 2025)
- Chair, Data Science Annual Evaluation Committee (Spring 2025)
- Member, College of Information, Administrator Evaluation Committee (Spring 2025)
- Member, Data Science Faculty Search Committee (Spring 2025)
- Member, Data Science Award Committee (Fall 2025, Sinha Scholarship)
- Member, Data Science Endowed Professor Review Committee (Fall 2025)
- Member, Data Science Executive Committee (Spring 2025 -)
- Member, Data Science Workload Committee (Spring, 2025)
- Co-Chair, Department of Information Science Reappointment, Promotion and Tenure Committee (Fall, 2024)
- Chair, Data Science Curriculum Work Group (Fall, 2024)
- Member, Department of Information Science Faculty Personnel Affairs Committee (Fall, 2024)
- Mentor, Junior Faculty Mentorship (Fall, 2024)
- Member, Data Science Tenure-track Faculty Search Committee for three tenure track assistant professors (Spring, 2024)
- Member, Data Science Faculty Search Committee at UNT, 2023-2024
- Member, External Affairs & Physical Resources Committee at UNT, 2023-2024
- Member, UH Systems Vice Chancellor/Vice President for Research Search Committee, 2023
- Faculty Advisor, UHCL ACM Chapter, 2021 - 2023
- Member, UHCL President's outstanding faculty award committee, 2020
- Member, College of Science and Engineering (CSE) FDF Committee at UHCL, 2021- 2023
- Member, CSE Mentoring Committee at UHCL 2021- 2023
- Member, CSE Graduate Research Assistant Committee at UHCL, 2017-2018
- Member, CSE Committee of Computing, Library, Research Services at UHCL, 2019-2021
- Member, CSE Policy and Advisory Committee at UHCL, 2019-2021
- Member, Computer Science (CS) New Faculty Search Committee at UHCL, 2022
- Member, Data Science (DS) New Faculty Search Committee at UHCL, 2022
- Member, CS Promotion and Tenure Review Committee at UHCL, 2021
- Member, Computer Engineering (CE) Promotion and Tenure Review Committee at UHCL, 2020
- Member, DS Committee at UHCL, 2020- 2023
- Member, DS Admission Committee at UHCL, 2020- 2023
- Member, DS Waiver Committee at UHCL, 2020 - 2023

- Member, CS Cybersecurity Committee at UHCL, 2017-2023
- Member, Writing Committee of Computing Sciences at UHCL, 2017-2019
- Member, CS Course Committee at UHCL, 2016-2023

Courses development and teaching at UNT

- Re-designed PhD in Data Science Program by adding 10+ new courses
- Re-designed MS in Data Science Program by adding 10+ new courses
- Re-designed DSTC 5502, Fall 2026
- Taught: INFO 5810 Fall 2023, 2024, 2025, Spring 2024, 2025, 2026; DTSC 5502, Fall 2026

Ph.D. Student supervised as Major/co-Major at UNT

Hanzhi Zhang (co-advised with Dr. Yunhe Feng), Sep 2023 –
Yuhan Zhou (co-advised with Dr. Haihua Chen), Sep 2024 –
Komala Subramanyam Cherukuri, Jan 2025 –
Zexuan Meng (co-advised with Dr. Yunhe Feng), Sep 2025 –
Sina Montazeri (co-advised with Dr. Yunhe Feng), Sep 2025 –
Anuradha Chandrasekaran (co-advised with Dr. Brady Lund), Sep 2025 –
Xuejie Chen (co-advised with Dr. Junhua Ding), Sep 2026 –
Arun Kadambi, Sep 2026 –
Gaurav Shekhar, Sep 2026 –

Ph.D. Committee Member

Huyen Thi Ngoc Nguyen (UNT)
Raef Abdallah (Wayne State University)